

Junyoung Park

PhD Applicant

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RESEARCH INTERESTS

My research lies at the intersection of structural dynamics, IoT sensing, and applied machine learning, with a focus on continuous monitoring and load-carrying capacity assessment of civil infrastructure. I develop high-resolution IoT sensors for long-term acceleration and strain measurements, cloud-based data pipelines for large-scale data management, and learning-based models that translate raw measurements into actionable indicators such as natural frequencies, displacement via data fusion, and vehicle load events. Beyond bridges, this approach extends to prefabricated structures during transportation and to port infrastructures, with the broader goal of enabling data-driven, proactive maintenance of safety-critical structures.

EDUCATION

Chung-Ang University, Seoul, Republic of Korea Mar 2022 – Feb 2024

M.S. in Civil Engineering (Structural Engineering)

- Thesis: A Scalable Bridge Health Monitoring System using an IoT Sensor and Cloud Computing
- Advisor: Prof. Jong-Woong Park · GPA 4.39 / 4.5

Chung-Ang University, Seoul, Republic of Korea Mar 2017 – Feb 2022

B.E. in Civil Engineering & B.S. in Mathematics (Double Degree)

- Thesis: Development of Low-power IoT Sensor and a Cloud-based Data Fusion Displacement Estimation Method for Ambient Bridge Monitoring
- GPA 4.20 / 4.5

RESEARCH EXPERIENCE

UDNS Co., Ltd., R&D Division, Yongin, Republic of Korea Apr 2024 – Present

Researcher (Industrial Technical Personnel Program)

- Designed signal-acquisition PCBs and developed weight-estimation algorithms for a strip-type High-Speed Weigh-in-Motion (HS-WIM) system; conducted indoor and field noise tests to validate signal-processing performance.
- Led the certification of an On-Board Mass (OBM) device for the Australian TCA (Transport Certification Australia) Category C scheme: analyzed certification requirements, prepared technical documentation, organized environmental, EMC, and IP-rated tests, managed accredited testing, and authored the final certification report.
- Served as the lead researcher on multiple Ministry of Land, Infrastructure and Transport (MOLIT) R&D projects; prepared interim and final reports, coordinated cross-institutional deliverables, and produced 2 journal papers, 3 patent applications, 3 software registrations, and 4 accredited test certificates.
- Authored R&D proposals for MOLIT, Ministry of the Interior and Safety, and related public agencies (3 of 4 proposals selected for funding).
- Led the renewal review for the Korean Excellent Logistics Technology (No. 2) designation, including application documents, presentation materials, and on-site review response.

Smart Infrastructure Technology Laboratory, Chung-Ang University Jun 2019 – Feb 2024

Graduate Researcher (Advisor: Prof. Jong-Woong Park)

Smart Bridge using IoT Sensors and Cloud-based Analytics

- Designed a stand-alone, event-driven IoT sensor for long-term acceleration and multi-channel strain monitoring with LTE data transmission to a cloud server.
- Developed a bridge data-management framework based on MySQL and InfluxDB, and an analysis pipeline for substructure condition evaluation.

Prefab Structure Monitoring from Fabrication to Construction

- Developed an ultra-low-power, multi-channel strain visualization sensor with E-paper QR display for whole-lifecycle monitoring of prefabricated structures.
- Implemented a KakaoTalk-based QR data acquisition pipeline coupled with automated analysis reporting via the ChatGPT API.

Prefab Structure Monitoring during Transportation

- Developed high-resolution IoT sensors for simultaneous acceleration and strain monitoring during transportation, with a BLE-to-LTE gateway for 1:N data aggregation.
- Integrated an automated analysis system covering lifting, transportation, and loading events.

Smart Maintenance for Port Structures

- Developed impact-detection and long-term behavior monitoring sensors for ship berthing/unberthing events.
- Designed an MQTT-based real-time data pipeline with InfluxDB storage.

IoT-based Smart Earthquake Detection · Smart Ultra-High-Performance Concrete

- Earlier projects (2019–2020): seismic-event IoT sensor with two-stage STA/LTA + frequency-based detection; smart UHPC self-stress-sensing circuit with 1:N beacon-based data collection.

JOURNAL PUBLICATIONS

1. **Park, J.**, Shin, J., & Won, J. Development of a Portable Multimetric Sensing System for Bridge Load Testing and Load-Carrying Capacity Evaluation. *Measurement* (under review), 2026.
2. **Park, J.**, Kim, J., Cho, Y., & Jung, Y. (2026). YOLO-Based Automatic Load Segment Detection for High-Speed Weigh-in-Motion Systems. *Journal of the Korean Society of Civil Engineers*, 46(1), 95–104.
3. Shin, J., **Park, J.**, Won, J., Park, J. W., & Min, J. (2024). Development of AI-Based Multisensory System for Monitoring Quay Wall Events. *Journal of Marine Science and Engineering*, 12(11), 1902.
4. **Park, J.**, Kim, J., Kim, J., & Cho, Y. (2024). Stabilization of Dynamic Weight Estimation Systems for Commercial Vehicles during Driving. *Journal of the Korean Society of Mechanical Technology*, 26(5), 1041–1046.
5. Khayam, S. U., Ajmal, A., **Park, J.**, Kim, I. H., & Park, J. W. (2023). Tendon Stress Estimation from Strain Data of a Bridge Girder Using Machine Learning-Based Surrogate Model. *Sensors*, 23(11), 5040.
6. Khayam, S. U., Won, J., Shin, J., **Park, J.**, & Park, J. W. (2023). Monitoring Precast Structures during Transportation Using a Portable Sensing System. *Automation in Construction*, 145, 104639.
7. **Park, J.**, Shin, J., & Park, J. W. (2022). Development of IoT Sensor and Cloud-Based Server for Bridge Long-Term Monitoring. *arXiv:2209.12186*.
8. Shin, J., **Park, J.**, & Park, J. W. (2022). Development of AI-Cloud Based High-Sensitivity Wireless Smart Sensor for Port Structure Monitoring. *arXiv:2209.13646*.
9. Won, J., Park, J. W., **Park, J.**, Shin, J., & Park, M. (2021). Development of a Reference-Free Indirect Bridge Displacement Sensing System. *Sensors*, 21(16), 5647.
10. Le, H. V., Kim, T. U., Khan, S., **Park, J. Y.**, Park, J. W., Kim, S. E., ... & Kim, D. J. (2021). Development of Low-Cost Wireless Sensing System for Smart Ultra-High Performance Concrete. *Sensors*, 21(19), 6386.
11. Khan, S., Won, J., Shin, J., **Park, J.**, Park, J. W., Kim, S. E., ... & Kim, D. J. (2021). SSVM: An Ultra-Low-Power Strain Sensing and Visualization Module for Long-Term Structural Health Monitoring. *Sensors*, 21(6), 2211.
12. **Park, J.**, Shin, J., Won, J., Park, J. W., & Park, M. (2021). Development of Low-Power IoT Sensor and Cloud-Based Data Fusion Displacement Estimation Method for Ambient Bridge Monitoring. *Journal of the Computational Structural Engineering Institute of Korea*, 34(5).
13. Won, J., **Park, J.**, Shin, J., Kang, S., Park, J. W., & Park, M. (2021). IoT Sensor and Cloud-Computing-Based Continuous Bridge Displacement Monitoring System Capable of Acceleration and Strain Sensing. *Magazine of the Korean Society of Civil Engineers*, 69(10), 58–65.
14. Won, J., **Park, J.**, Park, J. W., & Kim, I. H. (2020). BLESeis: Low-Cost IoT Sensor for Smart Earthquake Detection and Notification. *Sensors*, 20(10), 2963.

CONFERENCE PROCEEDINGS

1. **Park, J.**, Shin, J., Jang, H., & Park, J. W. (2022). Development of IoT Sensor System and Web-Based Document Automation for Bridge Load-Bearing Capacity Evaluation. *Proc. CONVR*.
2. **Park, J.**, Shin, J., & Park, J. W. (2023). Cloud-Based Ambient Bridge Monitoring using Strain and Acceleration Measurements. *Proc. ITC-CSCC*.

PATENTS

1. Park, J. W., **Park, J.**, & Shin, J. *Continuous Bridge Load-Carrying Capacity Assessment System and Method*. KR 10-2025-0174354. Application filed Nov 18, 2025.
2. Park, J. W., **Park, J.**, & Shin, J. *Automatic Load-Event Interval Recognition Using IoT Sensors for Structural Capacity Assessment*. KR 10-2853678. Registered Aug 28, 2025.
3. Park, J. W., **Park, J.**, & Kim, J. *Real-Time Dynamic Truck Operating Weight Correction Based on an FIR Filter*. KR 10-2025-0068214. Application filed Dec 30, 2024.
4. Park, J. W., **Park, J.**, & Shin, J. *Concrete Crack Trajectory Monitoring and Long-Term History Visualization*. KR 10-2024-0017869. Registered Feb 6, 2024.
5. Park, J. W., **Park, J.**, & Shin, J. *IoT Sensor-Based Integrated Monitoring for Precast Transportation*. KR 10-2022-0159449. Registered Nov 24, 2022.
6. Park, J. W., Won, J., **Park, J.**, & Shin, J. *Cloud-Based IoT Sensor Data Preprocessing and Deep-Learning Anomaly Detection*. KR 10-2022-0021506. Registered Feb 18, 2022.
7. Park, S., Shin, J., & **Park, J.** *Citizen-Participatory Smart City System Using Sensor Measurement and Data Visualization*. KR 10-2021-0126695. Registered Sep 24, 2021.
8. Won, J., **Park, J.**, & Park, J. W. *Smart IoT Earthquake Sensor Unit and Notification Method*. KR 10-2020-0059559. Registered May 19, 2020.

HONORS & AWARDS

Best Paper Presentation Award , Korea Institute for Structural Maintenance and Inspection	Apr 2024
Gold Award , X-Corps Plus Festival, National Research Foundation of Korea	Nov 2022
CAU Graduate Research Scholarship , full tuition, Chung-Ang University	Mar 2022
SeAH Haiam Scholarship , full undergraduate tuition (two years)	Jun 2020
Minister's Award , Smart Earthquake Detection Competition, Ministry of the Interior	Feb 2019
Grand Prize , AI Object Recognition Hackathon, 2018 Engineering Festival	Nov 2018

SKILLS

Programming: C, Python, MATLAB, Node.js, Swift

Engineering Tools: MATLAB/Simulink, Altium Designer, SolidWorks, PlatformIO, Xcode

Data & Infrastructure: MySQL, InfluxDB, MQTT, cloud (AWS), BLE/LTE embedded systems

Languages: English (TOEFL iBT 100/120, 2026), Korean (native)